

Technical Audit of Heart Disease Prediction Automated Decision System

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May, 6th 2025

1 Background

1.1 ADS Purpose

The purpose of this ADS is to predict whether an individual will have heart disease given the other factors (features) provided by the dataset. While the author does not state any specific goals for this particular ADS, the original dataset encourages that individuals apply machine learning methods to the data in order to detect patterns in the data that would allow us to identify the factors that have the greatest impact on heart disease. Thus, it is reasonable to believe that a primary objective behind the development of this ADS is to help with identification and prevention of these factors, as they would be crucial for healthcare and patient well being.

1.2 Potential Trade-Offs

The ADS' main goal is to identify which factors have the greatest impact on heart disease, maximizing for accuracy. This maximization of accuracy may ignore biases that are present in the data, as the ADS does not appear to take many precautions to prevent biases. This would result in a trade-off which exchanges fair treatment of those with heart disease for model accuracy.

2 Input and Output

2.1 Dataset Overview

The Automated Decision System (ADS) analyzed in this report utilizes the "Personal Key Indicators of Heart Disease" dataset available on Kaggle. This dataset contains health survey data collected by the Centers for Disease Control and Prevention (CDC) through their Behavioral Risk Factor Surveillance System (BRFSS) in 2020.

2.2 Data Collection Methodology

The BRFSS is a health-related telephone survey conducted annually by the CDC across all 50 states in the United States. The 2020 survey collected responses from over 400,000 adults regarding various aspects of their health status, including whether they have had coronary heart disease (CHD) or myocardial infarction (MI). The dataset used in the system represents a cleaned subset of this larger survey, focusing specifically on key indicators that may influence heart disease risk. The data collection followed a systematic telephone survey methodology, making it an observational dataset rather than data from a controlled clinical trial.

2.3 Feature Overview

The dataset contains 18 features, including demographic information, lifestyle factors, and health conditions that serve as potential predictors for heart disease. Table 1 provides an overview of these features, their data types, and descriptions.

Table 1: Dataset Features

Feature	Data Type	Description
HeartDisease	Binary categorical	Target variable indicating presence of heart disease
BMI	Numerical	Body Mass Index
Smoking	Binary categorical	Smoked at least 100 cigarettes in lifetime
AlcoholDrinking	Binary categorical	Heavy alcohol consumption status
Stroke	Binary categorical	History of stroke
PhysicalHealth	Numerical (0-30)	Days of poor physical health in past 30 days
MentalHealth	Numerical (0-30)	Days of poor mental health in past 30 days
DiffWalking	Binary categorical	Difficulty walking or climbing stairs
Sex	Binary categorical	Biological sex
AgeCategory	Ordinal categorical	Age range (13 categories from 18-24 to 80+)
Race	Categorical	Racial/ethnic group (6 categories)
Diabetic	Categorical	Diabetes status
PhysicalActivity	Binary categorical	Exercise in past 30 days
GenHealth	Ordinal categorical	Self-rated health status (5 levels)
SleepTime	Numerical	Average hours of sleep in 24-hour period
Asthma	Binary categorical	Asthma status
KidneyDisease	Binary categorical	Kidney disease status
SkinCancer	Binary categorical	Skin cancer status

2.4 Missing Values

The dataset is described as "cleaned" in its title, and analysis confirmed there are no missing values. This suggests that missing values were handled during a preprocessing stage before the dataset was made available for use.

2.5 Value Distribution

The data distribution across features provides critical insights into the underlying patterns of heart disease risk factors and demographic representation in the dataset. The target variable (HeartDisease) exhibits significant class imbalance, with positive cases substantially outnumbered by negative cases, a pattern that necessitates careful consideration during model training and evaluation to avoid bias toward the majority class. Demographic variables reveal important population characteristics, with age distribution showing higher representation of older adults (particularly those aged 65-74) as illustrated in Figure 3 in Appendix A, consistent with the age-related nature of cardiovascular disease. Gender distribution, visualized in Figure 4, demonstrates the well-documented higher prevalence of heart disease among males compared to females. Categorical health indicators present similarly informative patterns: BMI distribution (Figure 2) is left-skewed with Overweight (BMI 25-29.9) being the most common category comprising nearly one-third of the dataset, while General Health self-assessment (Figure 5) skews toward positive ratings with most individuals reporting "Very good" or "Good" health status, potentially reflecting a selection bias toward individuals who proactively engage with healthcare systems.

Racial representation within the dataset reveals notable disparities, with white individuals constituting approximately 162,295 records—more than all other racial groups combined—as shown in Figure 6, which raises important considerations about potential bias in model predictions and generalizability across diverse populations. Numerical features offer additional clinical insights: sleep patterns (Figure 7) show minimal differences between individuals with and without heart disease, echoing research linking both insufficient and excessive sleep to cardiovascular stress; physical and mental health metrics (Figure 8) demonstrate the expected association between heart disease and increased days of reported poor physical health; and mental health data reveals subtle connections between psychological wellbeing and cardiovascular outcomes. Lifestyle and comorbidity factors align with established medical knowledge, showing higher proportions of heart disease among respondents with history of stroke, diabetes (Figure 9), asthma, and kidney disease

(Figure 10)—conditions that share underlying vascular pathophysiology with coronary heart disease. These distribution patterns collectively provide a comprehensive view of the dataset’s composition and highlight potential areas where sampling limitations might influence model predictions, particularly with respect to demographic representation and class imbalance.

2.6 Feature Correlations

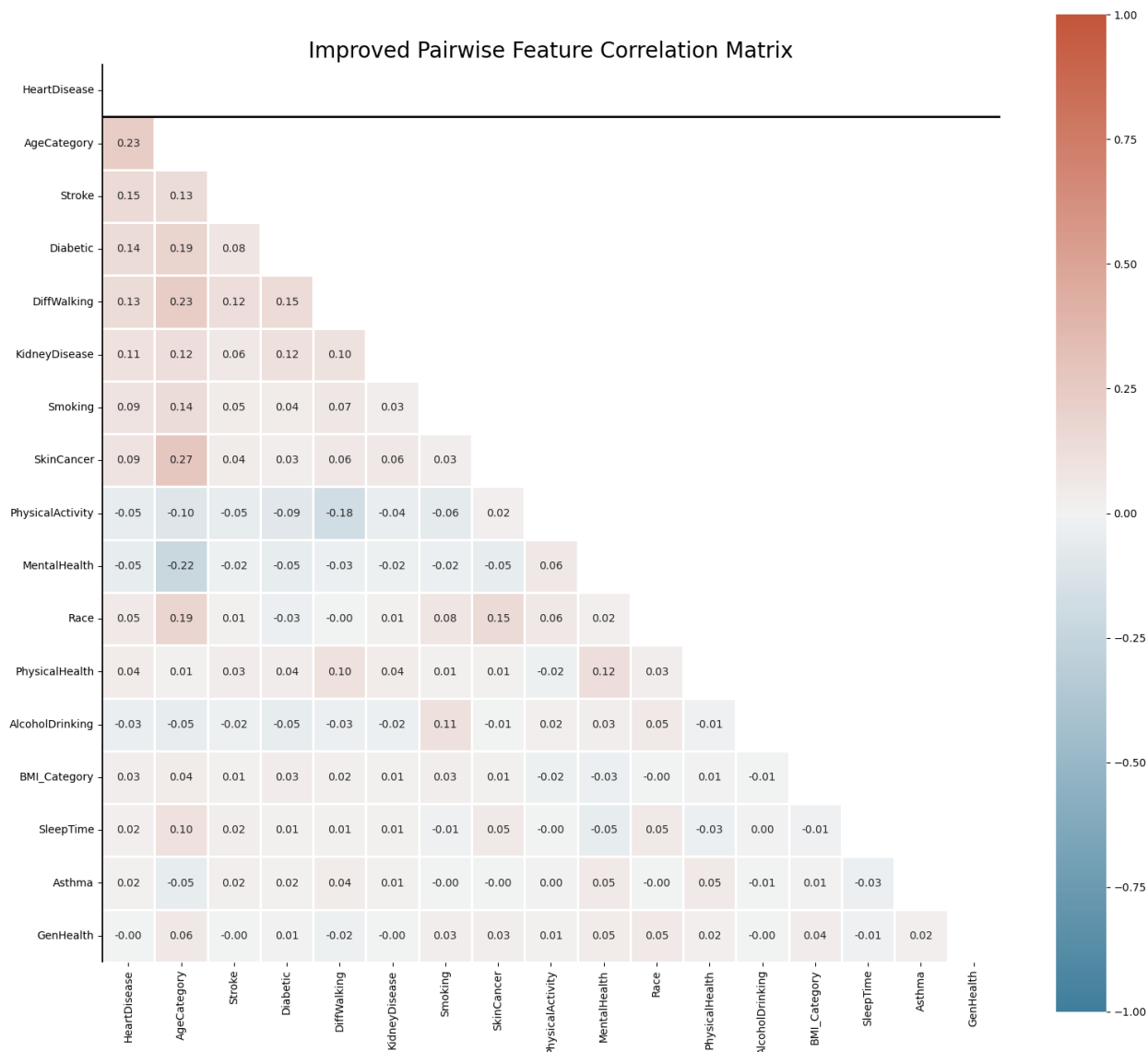


Figure 1: Pairwise Feature Correlation Matrix. Features are ordered by their absolute correlation with **HeartDisease**.

The pairwise correlation matrix illustrated in Figure 1 reveals important relationships between features in the heart disease dataset, with variables ordered by their absolute correlation strength with the target variable. Age emerges as the most influential predictor, exhibiting the strongest positive correlation with heart disease ($r = 0.23$), confirming the well-established clinical understanding that cardiovascular risk increases substantially with advancing age. Several comorbid conditions demonstrate moderate positive correlations, notably difficulty walking ($r = 0.13$), prior stroke history ($r = 0.15$), diabetic status ($r = 0.14$),

and kidney disease ($r \approx 0.11$), all of which share underlying vascular pathophysiological mechanisms that contribute to both heart disease and these related conditions.

Lifestyle and behavioral factors show more nuanced correlations with the heart disease outcome. Smoking and skin cancer history display modest positive associations ($r \approx 0.09$), with smoking’s moderate correlation potentially reflecting both its established causative role in heart disease and the tendency for smoking habits to change following diagnosis. Conversely, protective factors exhibit expected negative correlations, with physical activity, self-reported general health, and mental health status all showing weak inverse relationships with heart disease ($r \approx -0.05$), aligning with established evidence on the cardiovascular benefits of regular exercise and overall wellbeing.

The correlation matrix further reveals that most feature pairs exhibit relatively low absolute correlation values ($|r| < 0.10$), suggesting minimal multicollinearity among predictors. This limited feature interdependence indicates that most demographic and behavioral variables contribute independently to the predictive models rather than capturing redundant information. Such independence among predictors enhances model interpretability and suggests that the feature set effectively captures distinct aspects of cardiovascular risk. The correlation patterns observed provide valuable insights for both model development and clinical interpretation, highlighting key risk factors while confirming established medical understanding of heart disease etiology.

2.7 Output Description and Interpretation

The system produces a binary classification output indicating whether an individual is predicted to have heart disease ("Yes") or not ("No"). While the final output is binary, the underlying machine learning models calculate a probability score between 0 and 1, representing the likelihood of heart disease based on the input features. This probabilistic approach allows for more nuanced risk assessment beyond the simple binary classification, providing valuable information about prediction confidence that can be used in clinical decision-making. The probability scores enable healthcare providers to prioritize patients and allocate resources more effectively based on predicted risk levels rather than relying solely on categorical outcomes.

In clinical settings, prediction interpretation requires careful consideration of context and consequence. A prediction of "Yes" (probability ≥ 0.5) indicates that an individual is classified as having or being at high risk of heart disease based on their health indicators, potentially triggering additional diagnostic testing or preventive interventions. Conversely, a prediction of "No" (probability < 0.5) suggests lower cardiovascular risk, though it does not definitively rule out heart disease. It is crucial to recognize that such predictions serve as one component in a broader risk assessment process rather than as a definitive diagnosis, complementing rather than replacing clinical judgment, patient history, and standard diagnostic procedures in cardiovascular care.

The classification threshold represents a critical parameter that balances different types of errors and can be adjusted based on specific clinical priorities. The default threshold of 0.5 represents an arbitrary cutoff that may not be optimal for all clinical scenarios. Lowering the threshold increases sensitivity, potentially catching more true cases of heart disease at the expense of more false positives, which may be appropriate in screening contexts where missing cases carries serious consequences. Alternatively, raising the threshold increases specificity, reducing false alarms but potentially missing some true cases, which might be preferred in contexts where unnecessary testing or treatment has significant drawbacks. The optimal threshold ultimately depends on the relative costs of false positives versus false negatives in the specific clinical context, patient population characteristics, and available healthcare resources.

3 Implementation and Validation

The heart disease prediction system underwent a comprehensive data processing and model development pipeline to ensure robust performance. Data preprocessing began with rigorous outlier treatment utilizing the Interquartile Range (IQR) method, where values falling more than 1.5 IQR below the first quartile or above the third quartile were identified and removed from numerical features including BMI, PhysicalHealth, MentalHealth, and SleepTime. This process helped eliminate potential data anomalies that could have negatively impacted model accuracy and generalizability. Feature engineering enhanced the predictive power

of the dataset through thoughtful transformations, specifically converting the continuous BMI variable into a categorical feature with six standard medical classifications (Underweight, Normal weight, Overweight, Obesity I, II, and III) based on established clinical thresholds. Additionally, the Diabetic feature was simplified from four categories to two by combining borderline diabetes with negative cases and gestational diabetes with positive cases, streamlining the representation of this important cardiovascular risk factor.

The preparation of features for model training involved several key encoding techniques. Ordinal encoding was applied to ordered categorical features (BMI_Category, AgeCategory, Race, GenHealth), preserving the inherent ranking information within these variables. Nominal categorical features, particularly binary variables including HeartDisease and Smoking, were transformed using one-hot encoding to create appropriate numerical representations for the modeling algorithms. All numerical features underwent standard scaling to normalize their distributions to zero mean and unit variance, ensuring that variables with larger scales would not disproportionately influence model learning. While dimensionality reduction through Principal Component Analysis (PCA) was explored during development, it was ultimately not incorporated into the final models as it did not significantly improve performance while reducing interpretability. The class imbalance inherent in the dataset, with heart disease cases representing a minority class, was addressed using SMOTEENN (SMOTE with Edited Nearest Neighbors)—a hybrid approach that both oversamples the minority class and undersamples the majority class to create a more balanced training distribution without simply duplicating existing samples.

Five machine learning algorithms were evaluated for heart disease prediction, with XGBoost Classifier selected as the optimal model for implementation. XGBoost achieved the highest accuracy (90.3%) among all tested models, outperforming even other advanced ensemble methods like LightGBM (90.2%) and Random Forest (88.2%). The superior performance of XGBoost can likely be attributed to its ability to effectively capture the complex, non-linear relationships between health indicators and cardiovascular outcomes through gradient boosting and regularization techniques. This advanced ensemble learning approach enables the model to handle the inherent complexities of medical data, where risk factors often interact in sophisticated ways that simpler models like Logistic Regression (74.7%) cannot adequately represent.

Table 2: XGBoost Classifier Performance Metrics

Class	Precision	Recall	F1-score	Support
0.0 (No Heart Disease)	0.94	0.95	0.95	39,617
1.0 (Heart Disease)	0.32	0.27	0.29	3,175
Accuracy		0.90		42,792
Macro Average	0.63	0.61	0.62	42,792
Weighted Average	0.90	0.90	0.90	42,792

Further analysis of XGBoost’s performance metrics reveals important insights about its clinical applicability. As shown in Table 2, the model demonstrates excellent performance on the majority class (No Heart Disease) with precision, recall, and F1-scores all above 0.94. However, performance on the minority class (Heart Disease) is substantially lower, with precision of 0.32 and recall of 0.27. These metrics highlight a critical consideration in medical applications: while overall accuracy is high (0.90), the model’s ability to correctly identify positive heart disease cases requires careful consideration. This class-specific performance disparity is particularly important in healthcare contexts, where false negatives (missed heart disease cases) may have more serious consequences than false positives. The detailed performance metrics informed both model selection and potential clinical applications, acknowledging the differing implications of various error types in cardiovascular risk assessment.

4 Outcomes

4.1 Accuracy Analysis Across Subgroups

The performance of the ADS was evaluated across the different sex and racial groups to identify any disparities in prediction accuracy. This comparative analysis involved computing standard performance metrics

for the most accurate model, XGBoost, across these subpopulations.

	Male	Female
Recall	0.2703	0.2604
F1 Score	0.2897	0.2871
AUC	0.8013	0.8055

Table 3: Accuracy Metrics across Gender

	White	Black	Asian	Hispanic	American Indian/Alaskan Native	Other
Recall	0.2566	0.2984	0.2542	0.2769	0.3085	0.3000
F1 Score	0.2790	0.3263	0.2586	0.2951	0.3537	0.3349
AUC	0.8060	0.8128	0.7812	0.7833	0.8141	0.7971

Table 4: Accuracy Metrics across Race

4.2 Accuracy Analysis Across Subgroups

As shown in Table 3, model performance was consistent between male and female patients, with almost identical F1 scores (0.2897 for males and 0.2871 for females) and AUC values (0.8013 for males and 0.8055 for females). This indicates the model’s discriminative ability is equitable across sexes, though the low recall values (approximately 0.27 for both groups) highlight a general limitation in identifying positive heart disease cases regardless of sex.

Table 4 reveals more notable variations across racial groups. American Indian/Alaskan Native and Black populations showed the highest performance metrics, with AUC values of 0.8141 and 0.8128 respectively, and superior F1 scores of 0.3537 and 0.3263. Conversely, Asian patients had the lowest AUC (0.7812) and F1 score (0.2586), suggesting the model performs less effectively for this demographic. Hispanic patients also experienced relatively lower model performance with an AUC of 0.7833. These disparities, while not extreme, suggest that the model’s predictive capability may vary across racial groups, potentially reflecting underlying differences in how heart disease manifests or is diagnosed across different populations.

4.3 Fairness Analysis Across Subgroups

Our fairness evaluation across demographic groups revealed important patterns in the equitability of the XGBoost model’s predictions. Table 5 shows minimal fairness disparities between sexes, with nearly identical False Negative Rates (0.7396 for males and 0.7297 for females) and True Positive Rates (0.2604 for males and 0.2703 for females). Predictive parity was also similar between genders (0.3200 for males and 0.3121 for females), indicating that positive predictions carry similar clinical meaning regardless of sex.

However, Table 6 demonstrates more pronounced fairness disparities across racial groups. American Indian/Alaskan Native patients had the most favorable metrics, with the lowest False Negative Rate (0.6915) and highest True Positive Rate (0.3085), followed closely by Black patients. Most notably, predictive parity varied substantially, ranging from 0.2632 for Asian patients to 0.4143 for American Indian/Alaskan Native patients—a difference of over 15 percentage points. This indicates that a positive prediction for heart disease may carry different levels of reliability depending on a patient’s racial background. White and Asian populations showed the highest False Negative Rates (0.7434 and 0.7458 respectively), suggesting these groups face a greater risk of missed diagnoses. These disparities raise possible equity concerns about the model’s clinical application across diverse populations and may necessitate group-specific thresholds or additional screening protocols to ensure equitable care delivery.

4.4 Fairness Metrics Justification

The selected fairness metrics are believed to be appropriate in the healthcare context of heart disease prediction. Equal opportunity and minimized FNR disparity directly address the ethical imperative that all

	Male	Female
False Negative Rate	0.7396	0.7297
True Positive Rate	0.2604	0.2703
Predictive Parity	0.3200	0.3121

Table 5: Fairness Metrics Across Gender

	White	Black	Asian	Hispanic	American Indian/Alaskan Native	Other
False Negative Rate	0.7434	0.7016	0.7458	0.7231	0.6915	0.7000
True Positive Rate	0.2566	0.2984	0.2542	0.2769	0.3085	0.3000
Predictive Parity	0.3057	0.3598	0.2632	0.3160	0.4143	0.3789

Table 6: Fairness Metrics Across Race

patients, regardless of demographic characteristics, should have equal access to accurate diagnosis. In a clinical setting, missed diagnoses (false negatives) can delay treatment and lead to adverse health outcomes, making consistency in error rates across groups critically important. Predictive parity ensures that among those predicted to have heart disease, the proportion who actually have the disease is equal across groups. This is crucial to ensure that diagnostic predictions are equally reliable and trustworthy for all patients, allowing for fair treatment decisions. These fairness considerations align with broader healthcare equity principles that quality of care should not systematically vary based on patient demographics. Historical disparities in cardiovascular care and diagnosis across sex and racial groups make these fairness evaluations especially pertinent, as algorithmic systems may otherwise perpetuate or amplify existing inequities. By prioritizing these metrics, we acknowledge that in healthcare applications, certain types of errors carry significantly higher costs than others.

4.5 Additional Performance Analysis Methods

To gain deeper insights into our XGBoost model’s performance beyond standard accuracy metrics, we implemented two clinically relevant evaluation methods that address critical aspects of heart disease prediction systems.

4.5.1 Decision Boundary Analysis

We first analyzed the model’s performance on difficult cases—those where prediction probabilities fell near the classification threshold (between 0.4 and 0.6). These borderline cases represent patients for whom the model exhibits significant uncertainty and where clinical decision support would be most valuable.

Our analysis revealed that 1,565 instances (approximately 3.7% of the dataset) fell within this uncertainty region. While the model achieved 90.3% accuracy overall, performance on these boundary cases dropped substantially to just 54.2%—a reduction of 36.1 percentage points. This significant accuracy differential highlights a critical limitation: the model performs poorly precisely on the subpopulation where algorithmic guidance would be most beneficial in clinical practice.

This finding has important implications for deployment. In real-world clinical settings, healthcare providers would need to exercise particular caution when interpreting model predictions for patients with risk scores near the 0.5 threshold, potentially implementing additional screening or diagnostic procedures for these individuals.

4.5.2 Clinical Threshold Optimization

Given the asymmetric costs of different error types in cardiovascular care—where missed diagnoses (false negatives) can lead to life-threatening outcomes—we examined whether the standard 0.5 classification threshold maximizes clinical utility. Our threshold optimization analysis used a weighted utility function that prioritized sensitivity over specificity (3:1 ratio) to reflect the higher cost of missed heart disease cases.

The analysis identified 0.1 as the optimal threshold, yielding 63.7% sensitivity and 79.6% specificity. This substantially lower threshold increased the model’s ability to capture true heart disease cases, though at the cost of more false positives. With this threshold adjustment, the positive predictive value was 20.1%, indicating that approximately one in five positive predictions would be correct.

This threshold optimization represents an important clinical adaptation of the model, aligning its decision boundary with healthcare priorities that emphasize minimizing missed diagnoses. While this adjustment increases the rate of false positives, in a screening context for a serious condition like heart disease, this trade-off is clinically justified to ensure high-risk patients receive appropriate follow-up care.

5 Summary

5.1 Data Relevance to ADS

We believe that the data used is appropriate for this ADS because it is real world data collected by a credible organization (Centers for Disease Control and Prevention), and as a result is used in actual real world models for similar health evaluation purposes.

5.2 Implementation Assessment

For the XGBoost model audited, we believe that the implementation was robust and relatively fair. The nature of the XGBoost model allows it to handle imbalanced data and missing values well. While our accuracy and fairness analyses suggest some level of discrepancy across race groups, the results generally indicate that there is little to no bias for both gender and race. The model however could see improvement across certain accuracy metrics, especially in the ones measured in the accuracy analyses. The recall and F1 score values are notably low, indicating that the model frequently misses actual positive cases. Those with or at risk of heart disease would likely take great interest in the fairness and accuracy measures, as fair and accurate assessment of their health status by these models would be of positive benefit to their health. Other health-related bodies, such as hospitals and insurance companies, would likely find interest in the accuracy of the model, based on the various accuracy metrics. These metrics would allow them to determine individuals in need of their services.

5.3 Public Deployment of ADS

It is difficult to determine if the particular model (XGBoost model) analyzed from this ADS should be deployed in the public sector. While it demonstrates a high accuracy score, it is lacking in the other accuracy metrics. The model excels mostly at determining negative heart disease cases. As such, the model may only find practical use if deployed alongside other similar models that compensate for its lack of efficiency in detecting actual positive heart disease cases.

5.4 Possible Improvements to Methodology

The initial analysis conducted in the ADS does not consider the various subgroups, like the ones mentioned in our earlier analyses, that may be impacted by bias. It would be of interest to further analyze different fairness metrics aside from the ones mentioned earlier. It is also difficult to determine the legitimacy of the data processing done in the original dataset, as the ADS obtains this data from an outside source that claims to have included only the relevant variables.

6 Citations

Kamil Pytlak. (n.d.). Personal Key Indicators of Heart Disease [Data set]. Kaggle. Retrieved March 27, 2025, from <https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease/data>

Ebrahim Merza. (n.d.). Heart Disease Prediction — ML Acc 90 [Notebook]. Kaggle. Retrieved March 27, 2025, from <https://www.kaggle.com/code/ebrahimmerza/heart-disease-prediction-ml-acc-90>

A Feature Distributions

A.1 BMI Distribution

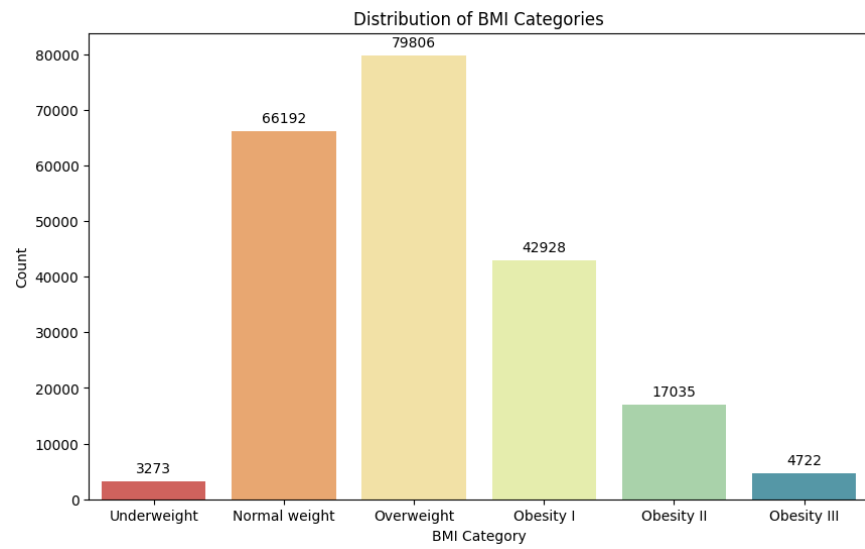


Figure 2: BMI Distribution

A.2 Age Distribution

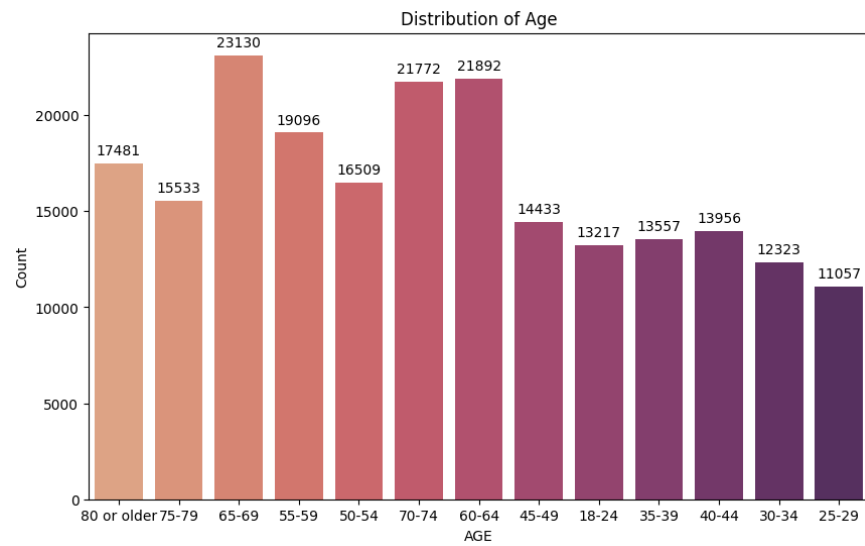


Figure 3: Age Distribution

A.3 Sex and Heart Disease

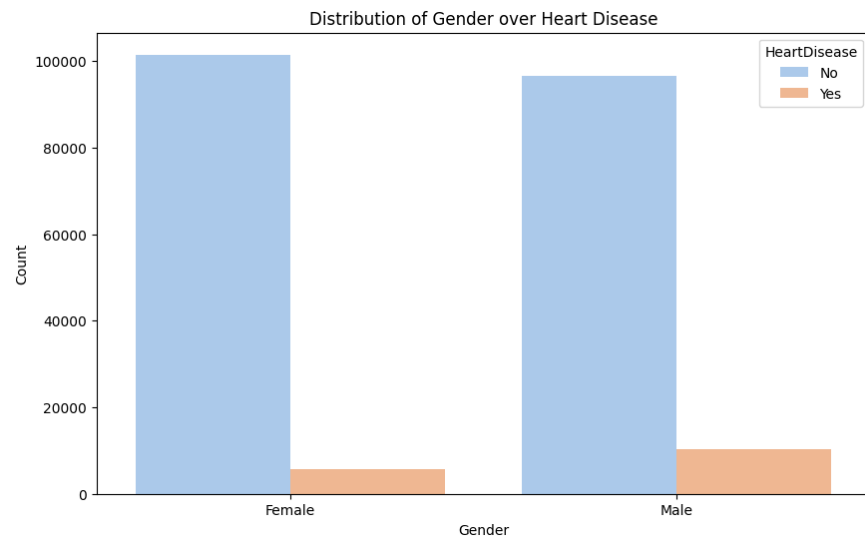


Figure 4: Heart Disease % by Sex

A.4 General Health Distribution

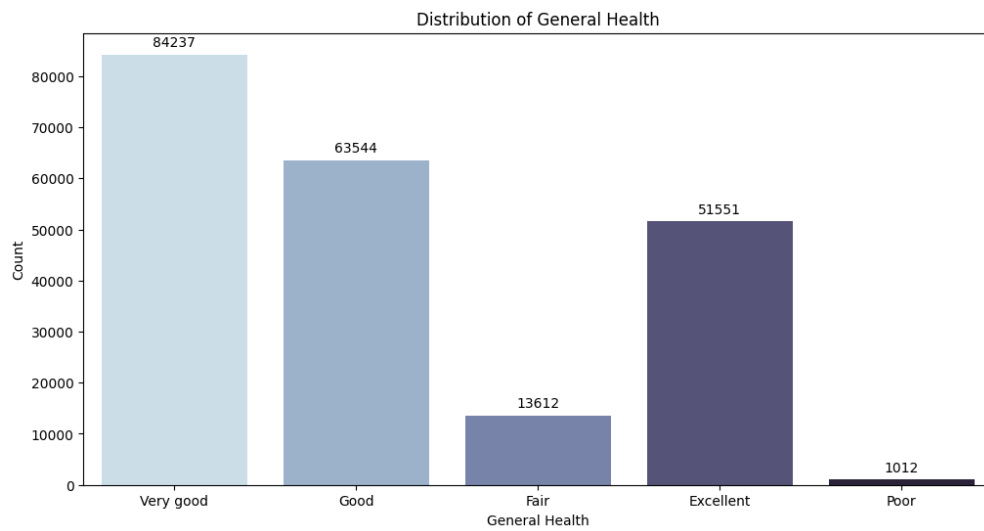


Figure 5: General Health Distribution

A.5 Race Distribution

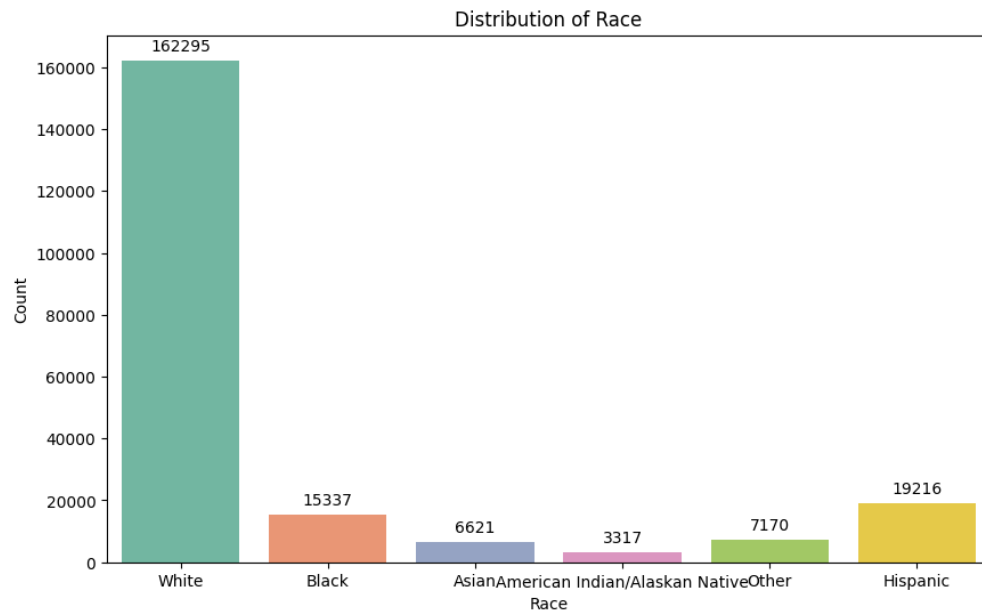


Figure 6: Race Distribution

A.6 Sleep Time and Heart Disease

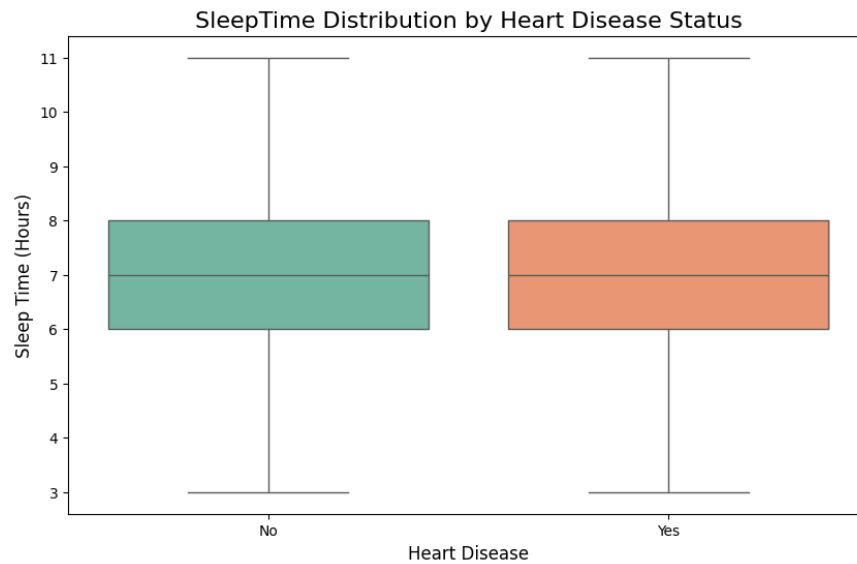


Figure 7: Heart Disease % by Sleep time

A.7 Physical and Mental Health

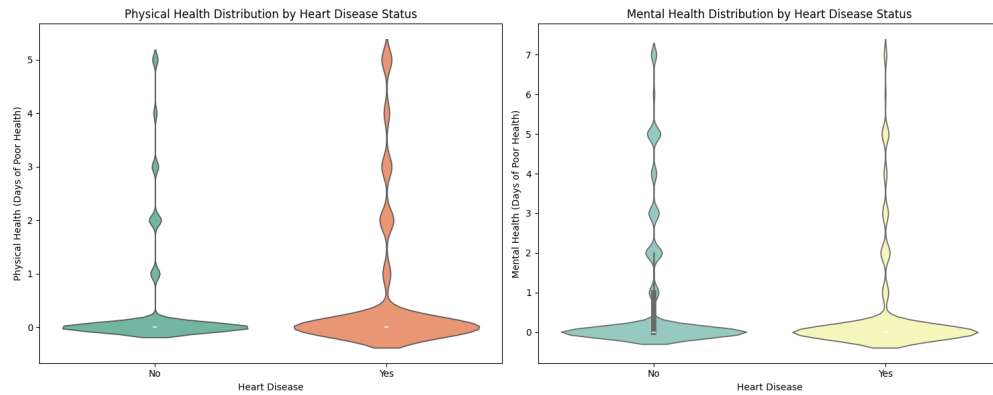


Figure 8: Heart Disease % by Physical and Mental health

A.8 Lifestyle Factors

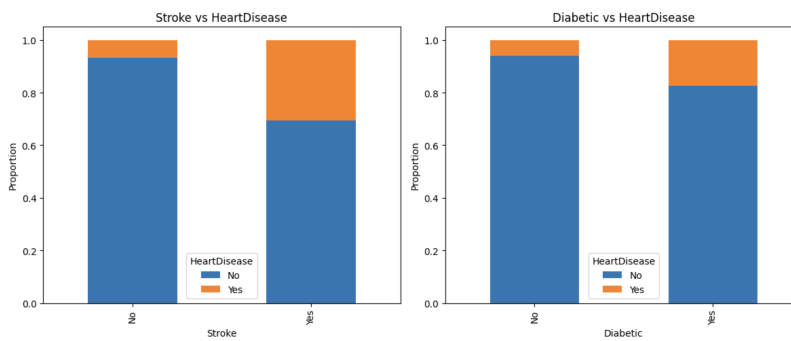


Figure 9: Stroke and Diabetes vs Heart disease

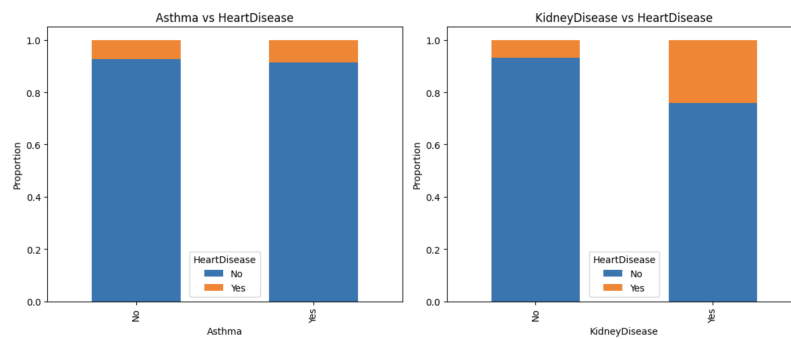


Figure 10: Asthma and Kidney Disease vs Heart disease